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4. Construct a scheduling program with C that selects the waiting process with the smallest execution time to execute next.

Code:

```
main.c
1 #include<stdio.h>
2 int main()
3 {
4     int bt[10],wt[10],tat[10],p[10];
5     int n,i,j,temp;
6     printf("Enter number of processes: ");
7     scanf("%d",&n);
8     for(i=0;i<n;i++)
9     {
10         p[i]=i+1;
11         printf("Enter Burst Time: ");
12         scanf("%d",&bt[i]);
13     }
14     for(i=0;i<n;i++){
15         for(j=i+1;j<n;j++)
16         {
17             if(bt[i]>bt[j])
18             {
19                 temp=bt[i];
20                 bt[i]=bt[j];
21                 bt[j]=temp;
22                 temp=p[i];
23                 p[i]=p[j];
24                 p[j]=temp;
25             }
26         }
27     }
28     wt[0]=0;
29     for(i=1;i<n;i++)
30     {
31         wt[i]=wt[i-1]+bt[i-1];
32         printf("\nProcess\tWT\tTAT\n");
33         for(i=0;i<n;i++)
34         {
35             tat[i]=wt[i]+bt[i];
36             printf("P%d\t%d\t%d\n",p[i],wt[i],tat[i]);
37         }
38     }
39     return 0;
40 }
```

Output:

```
Output
Enter number of processes: 3
Enter Burst Time: 6
Enter Burst Time: 2
Enter Burst Time: 8

Process WT  TAT
P2  0  2
P1  2  8
P3  8  16

=== Code Execution Successful ===
```